



CAT 2025

MBA RAPID REVISION

Lecture- 03

Revision Class – 3

Data Interpretation

By- GOURAV GUPTA



TOPICS *to be covered*

1) Basics DI

2)

3)



Set-1

Directions :



First 100 natural numbers are written on the board. First student entered the class and erased all numbers which were multiples of 2. Then second student entered and erased all multiples of 3 from the leftover numbers. Third student entered and erased all multiples of 4 from the leftover numbers. Similarly, every k^{th} student entered the class erases all the numbers which are multiples of ' $K+1$ '.

When all the students are inside the class, exactly 10 numbers are left on the board.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, - - - -

1st → 50

2nd → odd multi of 3 → 17

3rd → 4K → 0

4 → 5, 25, 35, 55, 65, 85, 95



#Q1. What is the difference between the largest and the smallest numbers remaining on the board?

$$44 \mid 96$$

#Q2. How many students are there in the class?

$$51 \mid 52 \mid 33 \mid 17 \mid 18 \mid 46$$

#Q3. What is the largest number remaining on the board?

$$97$$

#Q4. What is the sum of the numbers erased by the second student?

$$1275 \mid 1683 \mid 867 \mid 794$$

#Q5. What is the difference between the number of numbers erased by 5th and 6th students

$$7 \mid 4 \mid 1 \mid 10 \mid 3 \mid 2$$



1
59..
61..
67..
71..
73..
79..
83..
89..
97..

Last number to be erased? $\rightarrow$ 53

52nd $\rightarrow$ 53

#Q1. What is the difference between the largest and the smallest numbers remaining on the board?



$$97 - 1 = 96$$

#Q2. How many students are there in the class?



$52^{\text{nd}} \rightarrow 53$
 $53^{\text{rd}} \rightarrow X$
 $54^{\text{th}} \rightarrow X$
 $55^{\text{th}} \rightarrow X$
 $56^{\text{th}} \rightarrow X$
 $57^{\text{th}} \rightarrow$

$58^{\text{th}} \rightarrow 59 X$

#Q3. What is the largest number remaining on the board? → 97



#Q4. What is the sum of the numbers erased by the second student?

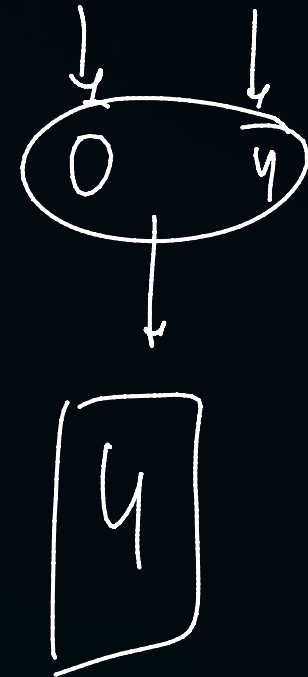


3, 9, 15, 21, - - - 99

$$3[289] = 867$$

#Q5. What is the difference between the number of numbers erased by 5th and 6th students

$$\begin{array}{l}
 1 \\
 2 \\
 3 \\
 4 \\
 \textcircled{5} \rightarrow 6K \rightarrow 0 \\
 \textcircled{6} \rightarrow 7K \rightarrow \left. \begin{array}{l} 7 \times 1 = 7 \\ 7 \times 7 = 49 \\ 7 \times 11 = 77 \\ 7 \times 13 = 91 \end{array} \right\} \textcircled{4}
 \end{array}$$

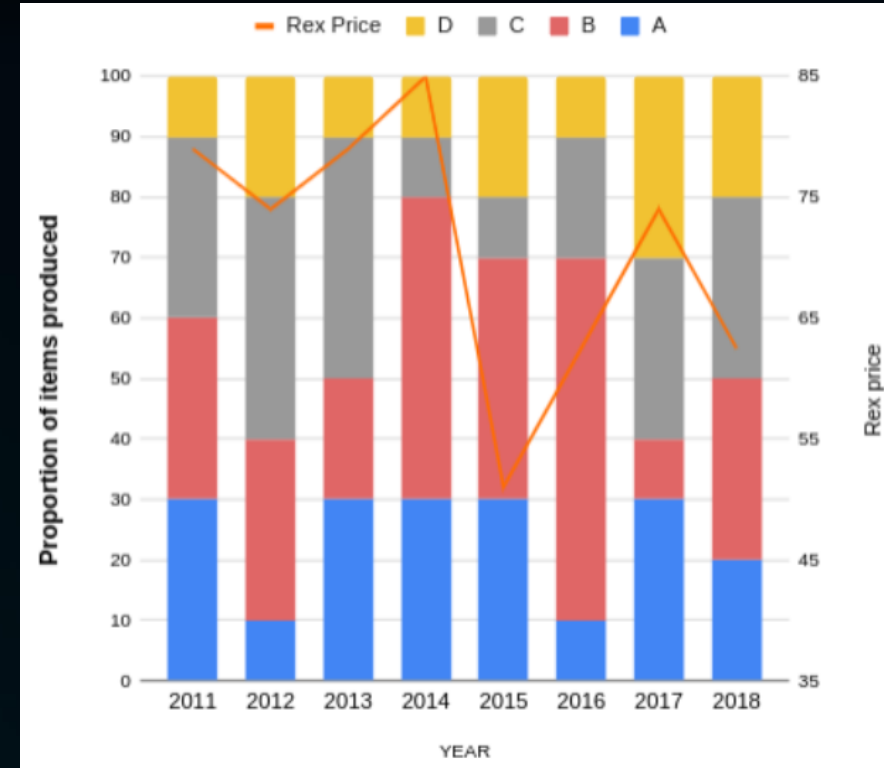




Set-2

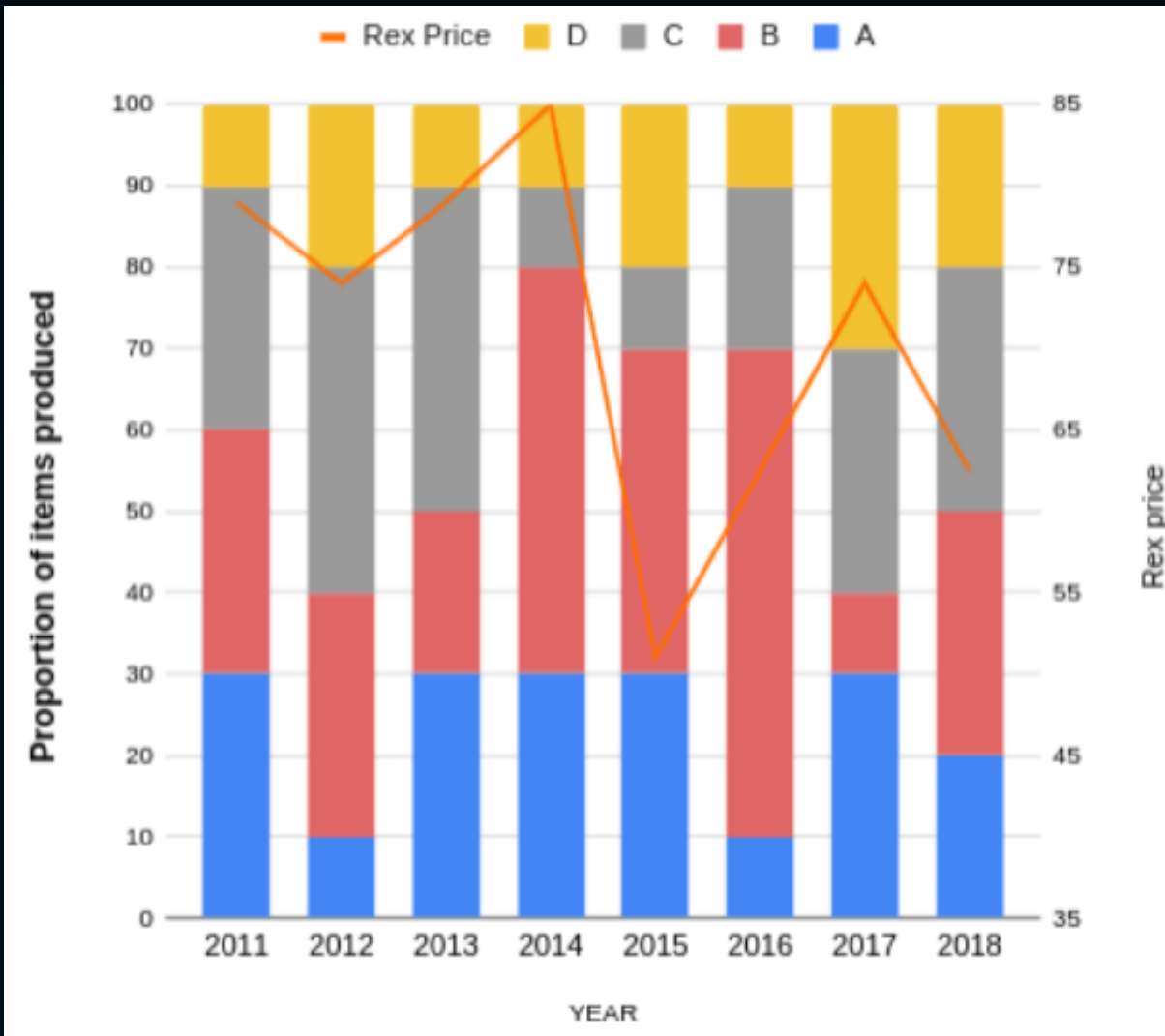
Directions :

Alpha company uses Rex as main raw material for manufacturing Item A, Item B, Item C and Item D. The graph below shows the proportion of items produced and price trend of Rex. Annual Res_Index of an item in a given year is measured as follows:



Year	2011	2012	2013	2014	2015	2016	2017	2018
Total Four Items Annual Production (Million Units)	1600	1000	1500	900	1250	2000	1000	1600

Res_Index of Item A in Year 2013 = Absolute Value (Production of Item A in 2013 – Production of Item A in 2012) / Absolute Value (Rex Price in 2013 – Rex Price in 2012)



Res_Index of Item A in Year 2013 = Absolute Value (Production of Item A in 2013 – Production of Item A in 2012) / Absolute Value (Rex Price in 2013 – Rex Price in 2012)

$$\text{Res Ind of A (2013)} = \frac{A_{2013} - A_{2012}}{R_{2013} - R_{2012}}$$

Year	2011	2012	2013	2014	2015	2016	2017	2018
Total Four Items Annual Production (Million Units)	1600	1000	1500	900	1250	2000	1000	1600

A. 2015 **B. 2016** **C. 2017** **D. 2018**

A. 2012 **B. 2013** **C. 2016** **D. 2017**

A. 9 **B. 8** **C. 7** **D. 6**

A. 2012 **B. 2018** **C. 2014** **D. 2017**

Questions :



#Q-1 In which year, Rex Price had highest percentage increase?

- A. 2015 ☒ B. 2016 ~~C. 2017~~ D. 2018

2015 → 2016

π

2016 → 2017

π



Year	2011	2012	2013	2014	2015	2016	2017	2018
Total Four Items Annual Production (Million Units)	1600	1000	1500	900	1250	2000	1000	1600

Questions :



#Q-2 From the years mentioned below, in which year average Res_Index of all items manufactured was highest?

A. 2012

B. 2013

C. 2016

D. 2017

	11	12	13	15	16	17
A	480	100	450	375	200	300
B	480	300	300	500	1200	100
C	480	400	600	125	400	300
D	160	200	150	250	200	300
Avr	79	74	79	51	62.5	74

$$\begin{aligned} \underline{2012} &= \frac{380}{5} + \frac{180}{5} + \frac{80}{5} + \frac{40}{5} \\ &= \frac{680}{5} = 136 \end{aligned}$$

$$\begin{aligned} \underline{2013} &= (350 + 0 + 200 + 50) / 5 \\ &= 600 / 5 = 120 \end{aligned}$$

$$\begin{aligned} \underline{2016} &= (175 + 700 + 275 + 50) / 11.5 \\ &= 1200 / 11.5 = 104.3 \end{aligned}$$

$$2017 = 1400 / 11.5 \quad \times$$



Questions :



#Q-3 How many times proportion of an Item manufactured has not been changed in the subsequent year?

- A. 9 B. 8 **C. 7** D. 6

A → 2
 B → 1
 C → 3
 D → 1

7



Year	2011	2012	2013	2014	2015	2016	2017	2018
Total Four Items Annual Production (Million Units)	1600	1000	1500	900	1250	2000	1000	1600

Questions :



#Q-4 From the years mentioned below, in which year Res_Index for item D was highest?

~~A. 2012~~

~~B. 2018~~

☒ C. 2014

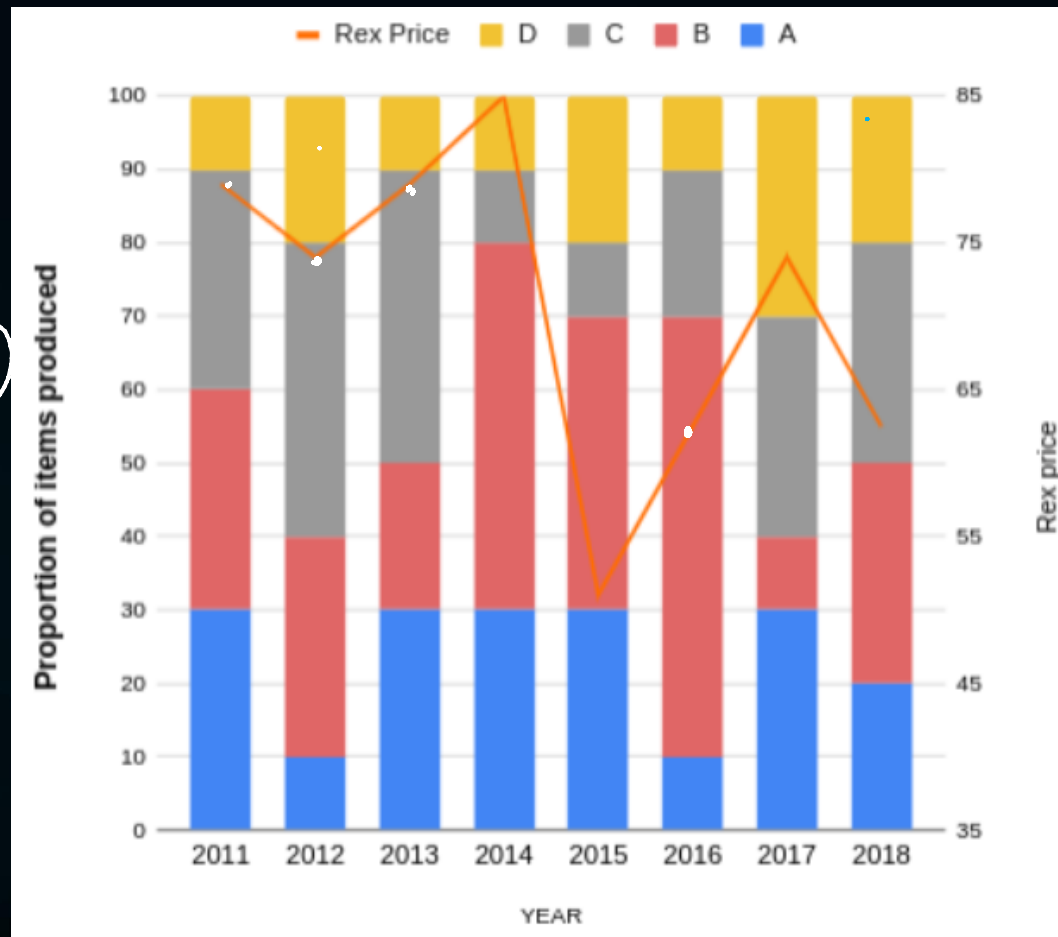
~~D. 2017~~

$$RI(2012) = \frac{|D_{2012} - D_{2011}|}{|R_{2012} - R_{2011}|} = \frac{200 - 160}{74 - 79} = \frac{40}{5} = 8$$

$$2014 = \frac{|190 - 150|}{|85 - 79|} = \frac{40}{6} = 10$$

$$2017 = \frac{300 - 200}{74 - 62.5} = \frac{100}{11.5} \times$$

$$\frac{320 - 300}{62.5 - 74} = \frac{20}{11.5} \times$$



Year	2011	2012	2013	2014	2015	2016	2017	2018
Total Four Items Annual Production (Million Units)	1600	1000	1500	900	1250	2000	1000	1600



Set-3

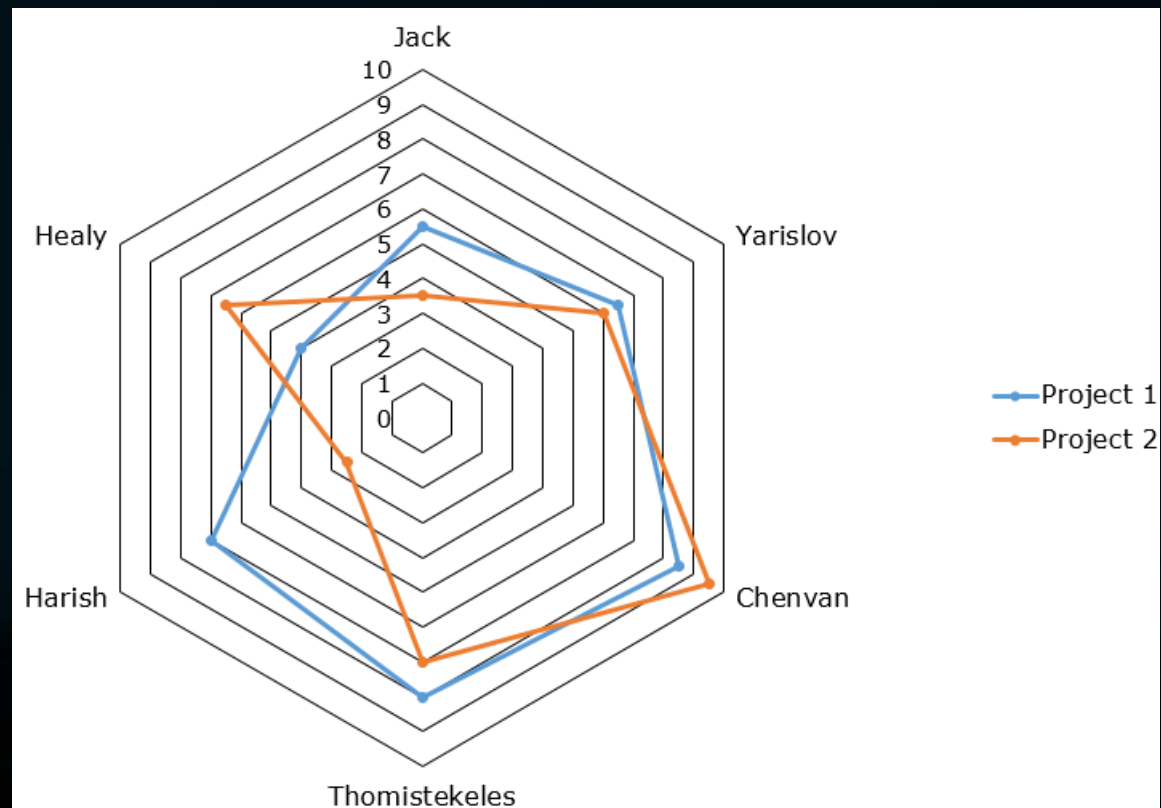
Read the following data and answer the given questions.



At Yokihama University, six foreign exchange students, i.e., Jack, Yarislov, Chenvan, Thomistekes, Harish, and Healy are given two projects, Project 1 and Project 2.

The projects are evaluated by a panel of faculties at Yokihama University and are rated from 1–10.

The ratings given by the panel are shown in the given radar chart.

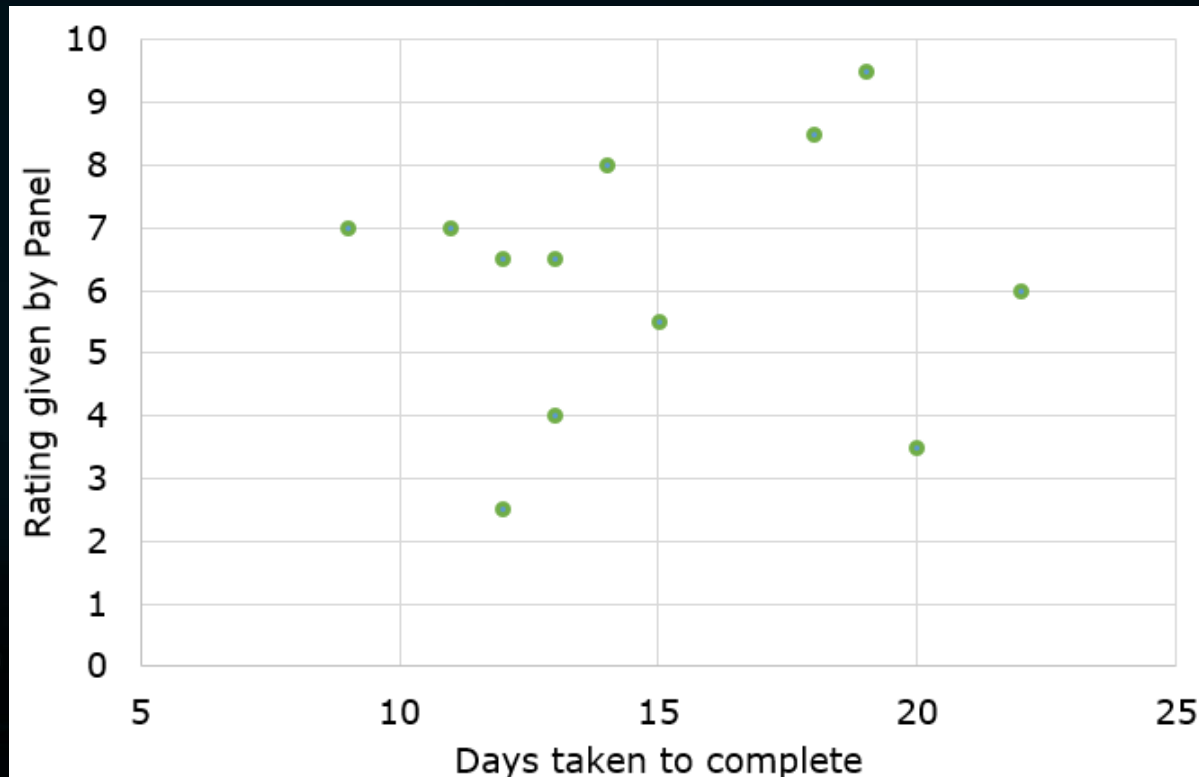


Read the following data and answer the given questions.



Hemaloan was given the responsibility to compile the days taken by 6 foreign exchange students to complete the two projects in two different charts. Hemaloan slacked and compiled the number of days taken by the six students of the two projects in the same chart without any distinction. The data that he compiled is given in the scatter chart given below.

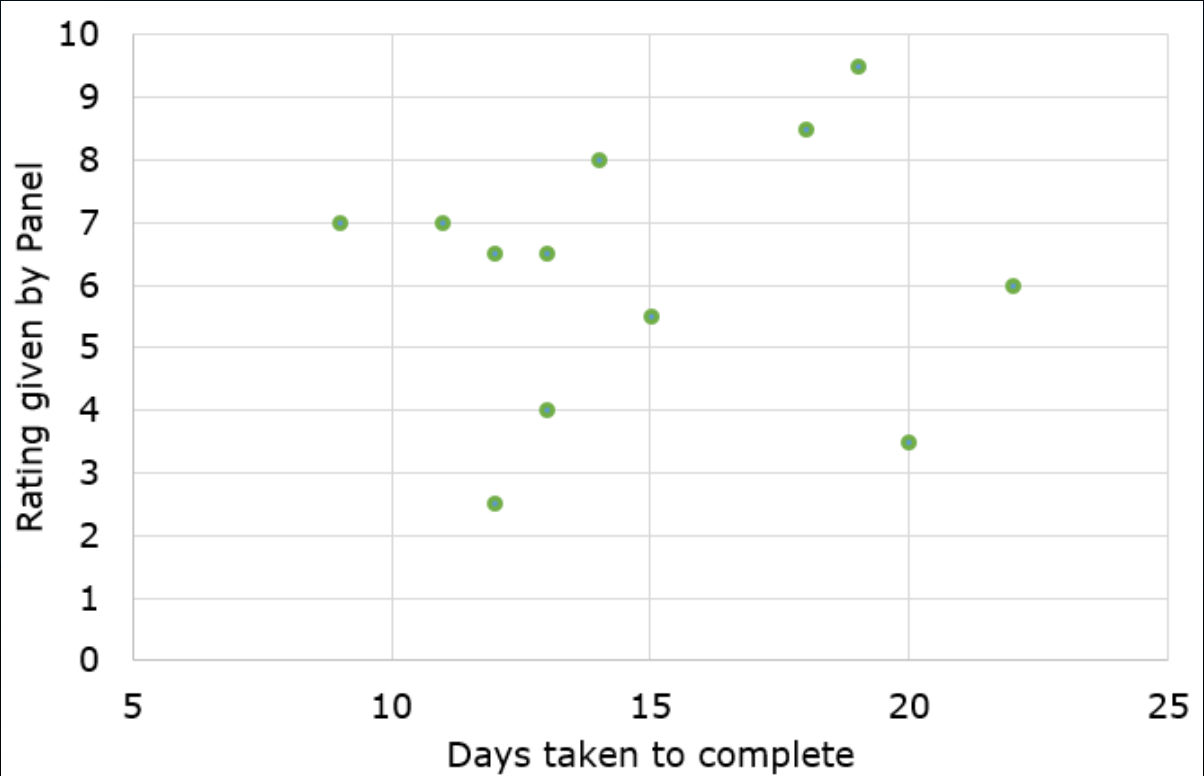
It is known that the difference between the days taken to complete projects 1 and 2 is less than 5 for four foreign exchange students. The rating of all the students is a multiple of 0.5.





At Yokihama University, six foreign exchange students, i.e., Jack, Yarislov, Chenvan, Thomistেকেles, Harish, and Healy are given two projects, Project 1 and Project 2. The projects are evaluated by a panel of faculties at Yokihama University and are rated from 1–10. The ratings given by the panel are shown in the given radar chart.

Hemaloan was given the responsibility to compile the days taken by 6 foreign exchange students to complete the two projects in two different charts. Hemaloan slacked and compiled the number of days taken by the six students of the two projects in the same chart without any distinction. The data that he compiled is given in the scatter chart given below. It is known that the difference between the days taken to complete projects 1 and 2 is less than 5 for four foreign exchange students. The rating of all the students is a multiple of 0.5.





#Q1. For how many of the 6 students, the days taken to complete both the projects can be uniquely determined?

#Q2. Which of the following is nearest to the average of the sum of rating of all the students who completed each of the given projects in less than 15 days?

A. 11 B. 12 C. 10 D. Cannot be determined

#Q3. For how many students, the days taken to complete the projects increased from Project 1 to Project 2 along with a rating change equal to or less than 1.0?

#Q4. For how many students, their rating was more than 5 in both the projects and the number of days taken by him were less than 18 in at least one of the projects?

Questions



#Q-1. For how many of the 6 students, the days taken to complete both the projects can be uniquely determined?

4

		Proj - 1	Proj - 2
✓	x J (5)	5.5 → 15	3.5 → 20
✗	x Y	6.5 → 12/13	6 → 22
✓	✓ C (1)	8.5 → 18	9.5 → 19
✓	T	8 → 14	7 → 11
✓	Har	7 → 9	2.5 → 12
✗	Hea	4 → 13	6.5 → 13/12

Questions



#Q-2. Which of the following is nearest to the average of the sum of rating of all the students who completed each of the given projects in less than 15 days?

A. 11

B. 12 ✓✓

C. 10

D. Cannot be determined ✗

$$\frac{15 + 9.5 + 10.5}{3}$$

$$\frac{35}{3} = 11.66$$

	Proj - 1	Proj - 2
✗ J (5)	5.5 → 15	3.5 → 20
✗ Y	6.5 → 12/13	6 → 22
✗ C (1)	8.5 → 18	9.5 → 19
✓ T	8 → 14	7 → 11
✓ Har	7 → 9	2.5 → 12
✓ Meer	4 → 13	6.5 → 13/12

Questions



#Q-3. For how many students, the days taken to complete the projects increased from Project 1 to Project 2 along with a rating change equal to or less than 1.0?

2

		Proj - 1	Proj - 2
X	J (5)	5.5 → 15	3.5 → 20
✓	Y	6.5 → 12/13	6 → 22
✓	C (1)	8.5 → 18	9.5 → 19
X	T	8 → 14	7 → 11
X	Har	7 → 9	2.5 → 12
X	Har	4 → 13	6.5 → 13/12

Questions



#Q-4. For how many students, their rating was more than 5 in both the projects and the number of days taken by him were less than 18 in at least one of the projects?

2

	Proj - 1	Proj - 2
X x J (5)	5.5 → 15	(3.5) → 20
✓ x Y	6.5 → 12/13	6 → 22
X ✓ C (1)	8.5 → 18	9.5 → 19
✓ T	8 → 14	7 → 11
X Har	7 → 9	(2.5) → 12
f Har	(4) → 13	6.5 → 13/12

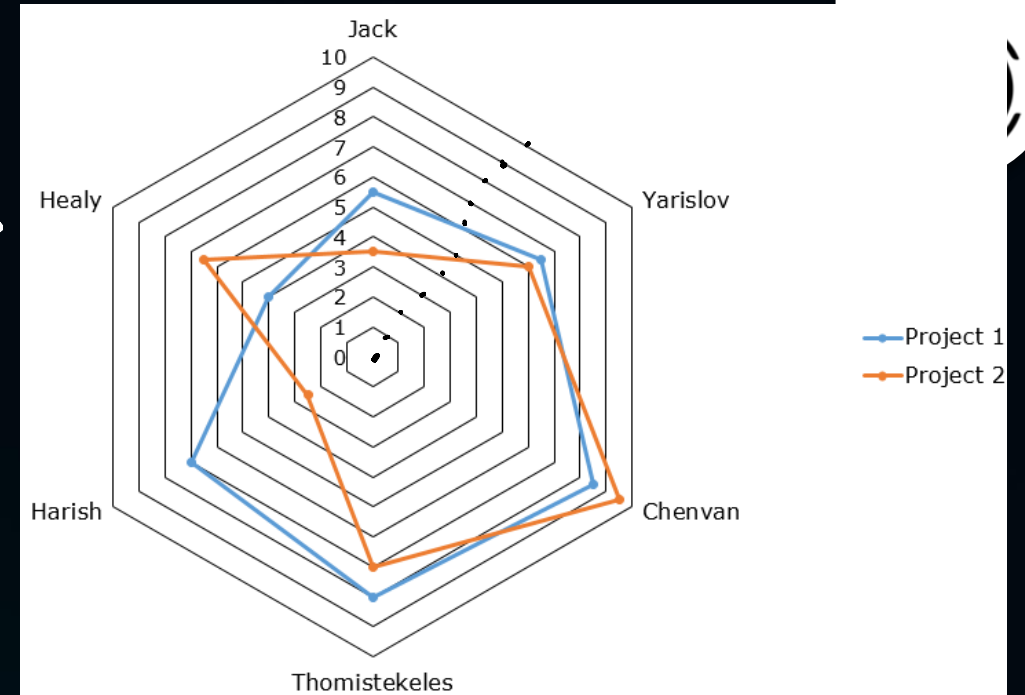
	Proj - 1	Proj - 2
x J (5)	5.5 → 15	3.5 → 20
x Y	6.5 → 12/13	6 → 22
✓ C (1)	8.5 → 18	9.5 → 19
T	8 → 14	7 → 11
Har	7 → 9	2.5 → 12
Hea	4 → 13	6.5 → 13/12

(6.5)

12 < 13

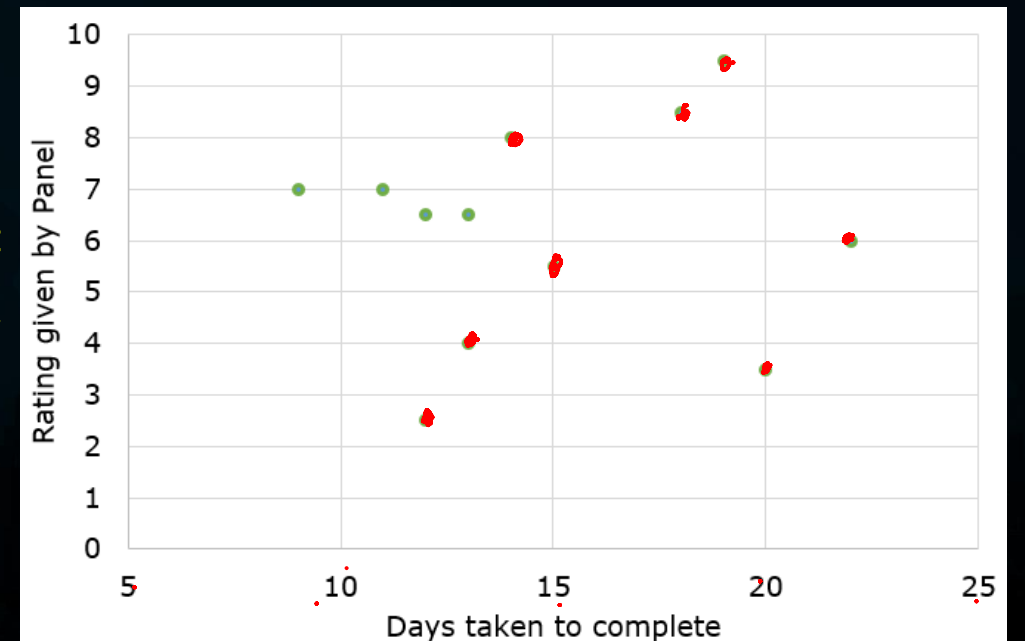
(7)

9 < 11



Hemaloan was given the responsibility to compile the days taken by 6 foreign exchange students to complete the two projects in two different charts. Hemaloan slacked and compiled the number of days taken by the six students of the two projects in the same chart without any distinction. The data that he compiled is given in the scatter chart given below.

It is known that the difference between the days taken to complete projects 1 and 2 is less than 5 for four foreign exchange students. The rating of all the students is a multiple of 0.5.





Thank You





THANK
You

